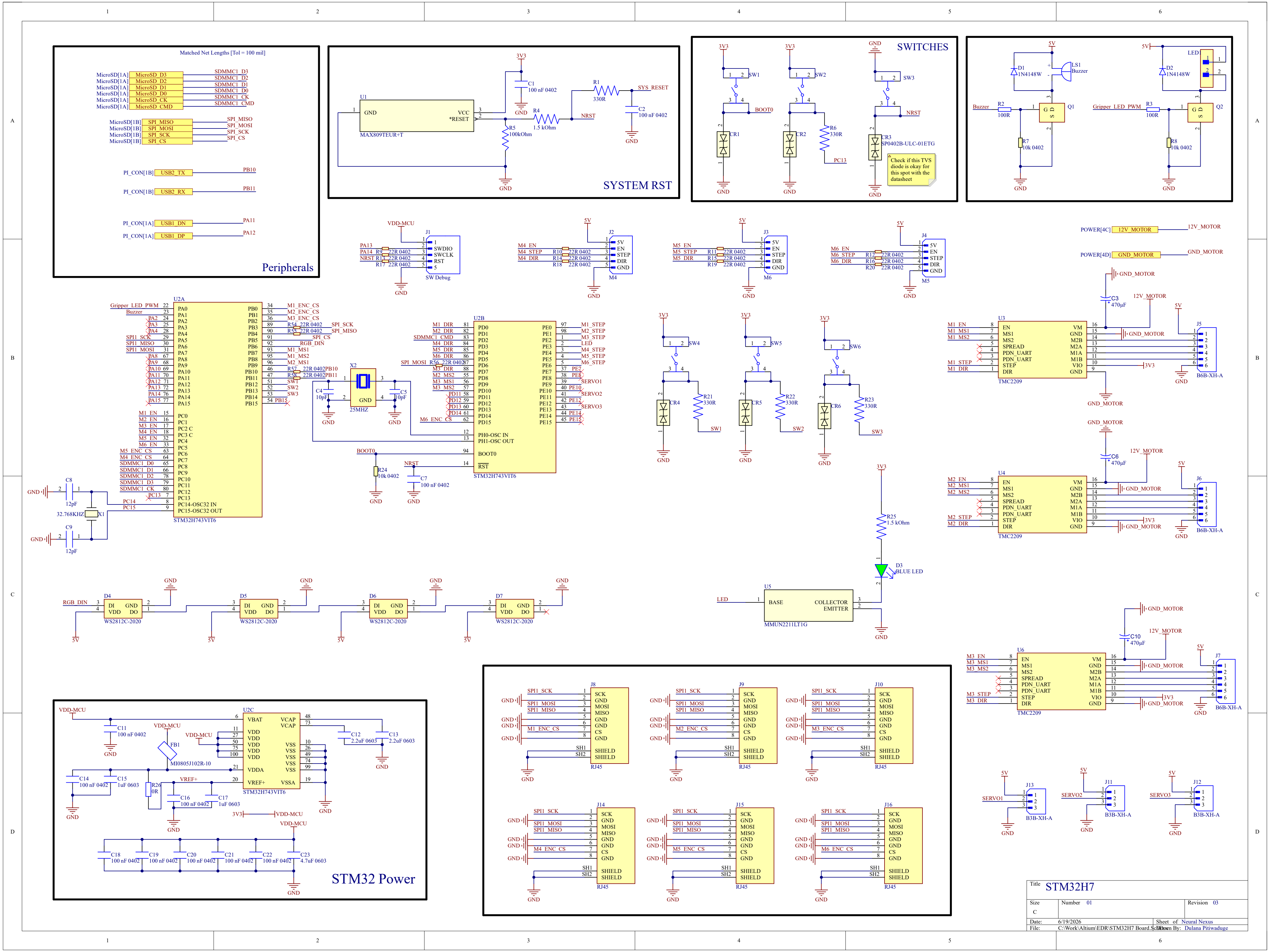
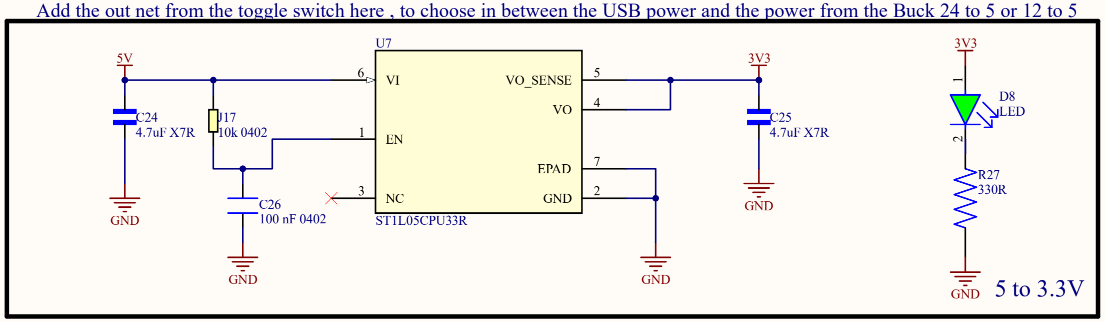


Matched Net Lengths [Tol = 100 mil]
Peripherals
SYSTEM RST
SWITCHES
STM32 Power
U2A STM32H743VIT6
U2B STM32H743VIT6
U2C STM32H743VIT6
U3 TMC2209
U4 TMC2209
U6 TMC2209
MAX809TEUR+T
MMUN2211LT1G
WS2812C-2020
SW Debug
RJ45
B6B-XH-A
B3B-XH-A
BLUE LED
25MHZ
32.768KHZ
Check if this TVS diode is okay for this spot with the datasheet
Title STM32H7
Size C
Number 01
Revision 03
Date: 6/19/2026
Sheet of
Neural Nexus
File: C:\Work\Altium\EDR\STM32H7 Board.SchDoc
Drawn By: Dulana Pitiwaduge

Add the out net from the toggle switch here , to choose in between the USB power and the power from the Buck 24 to 5 or 12 to 5



Splitting the 12V line is only half the battle; you must also manage the ground return paths.

When those three motors pull 6A, that 6A has to flow back to the battery through the ground plane. If that 6A flows underneath or through the ground copper near your STM32 or your TPS56637 feedback resistors, it will raise the local ground voltage, causing your microcontroller to read false button presses, corrupt SPI data, or trigger a brown-out reset.

How to route it in Altium:

Do not use one massive, uniform ground pour for the whole board.

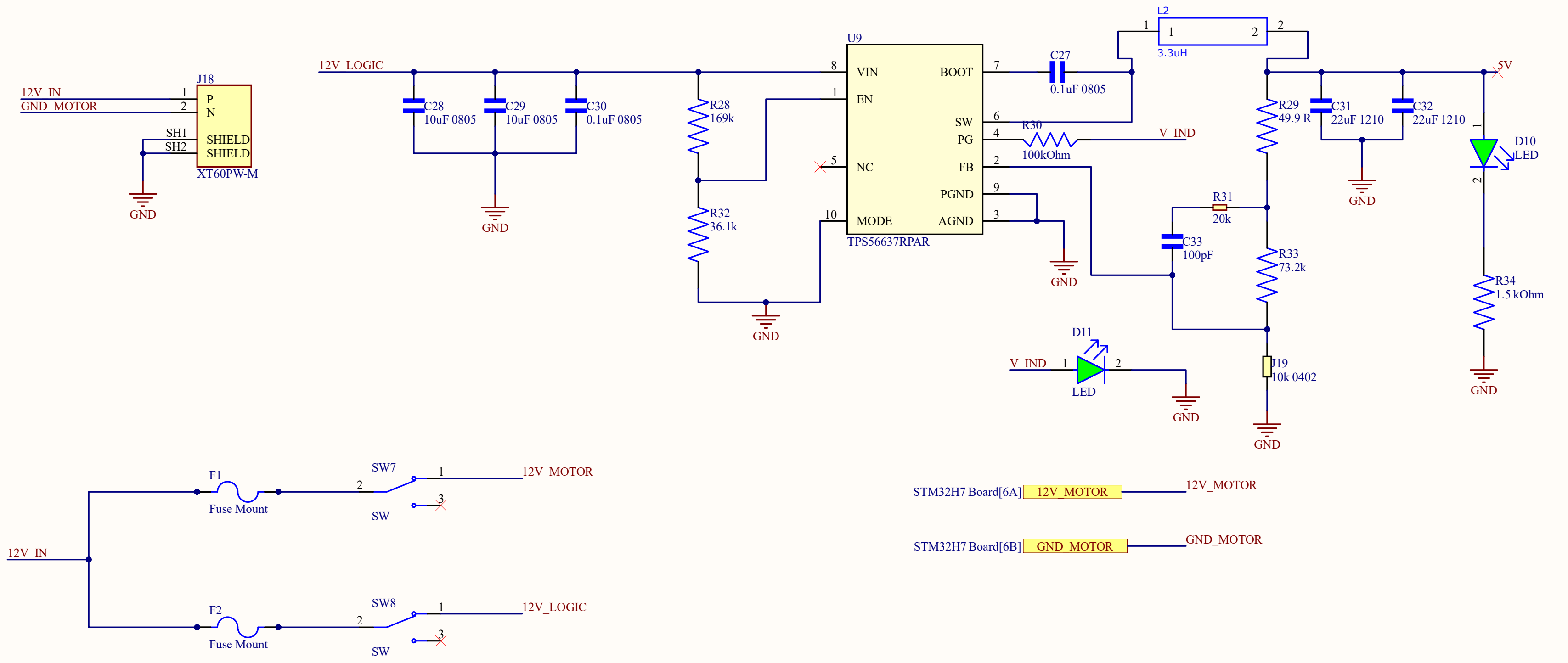
Create a specific, thick ground trace (or dedicated polygon pour) that connects the GND pins of the three TMC2209s directly back to the PGND pin of the XT60 connector.

Keep this motor ground return path physically away from your STM32 and analog sensors.

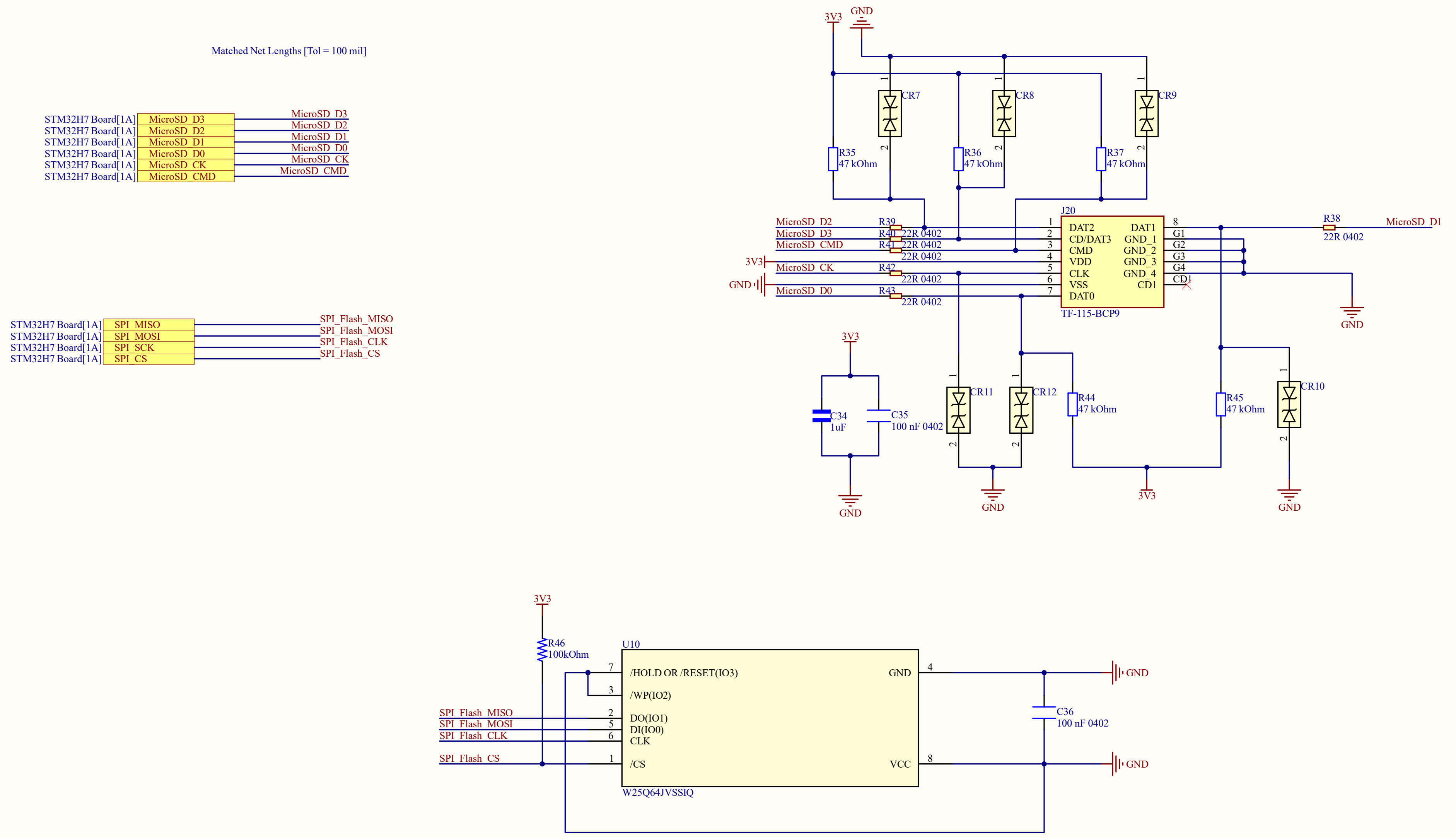
Your logic ground (PGND and AGND from the buck converter, and the STM32 ground) should meet the motor ground at exactly one single point: the physical solder pad of the XT60's negative pin.

3. Don't Forget the Bulk Capacitors!

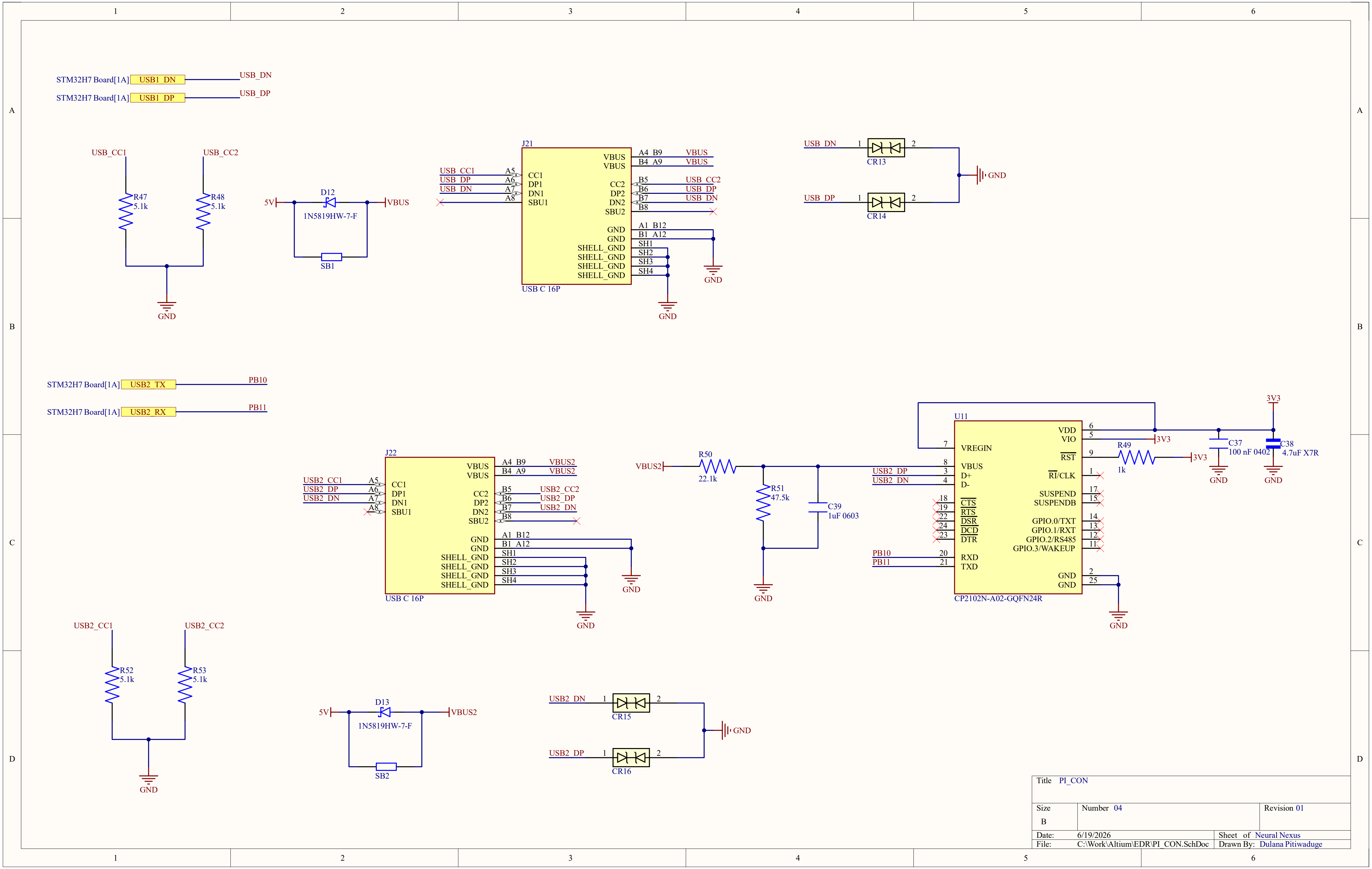
As a final reminder, because you are running 3 motors from this 12V line, you absolutely must place a large aluminum electrolytic capacitor (e.g., 220uF to 470uF, rated for 35V or 50V) on the 12V_MOTORS net right next to the TMC2209 drivers. Without it, the inductive flyback from three simultaneous 2A motors will easily spike the 12V rail past 30V and detonate your chips.



Title	POWER		
Size B	Number 02		Revision 04
Date:	6/19/2026	Sheet of	Neural Nexus
File:	C:\Work\Altium\EDR\POWER.SchDoc	Drawn By:	Dulana Pitiwaduge



Title	MicroSD & SPI FLASH		
Size B	Number 03		Revision 04
Date:	6/19/2026	Sheet of	Neural Nexus
File:	C:\Work\Altium\EDR\MicroSD.SchDoc	Drawn By:	Dulana Pitiwaduge



STM32H7 Board[1A] USB1_DN USB_DN
STM32H7 Board[1A] USB1_DP USB_DP
USB_CC1 R47 5.1k
USB_CC2 R48 5.1k
GND
5V D12 1N5819HW-7-F VBUS
SB1
J21
USB_CC1 A5 CC1
USB_DP A6 DP1
USB_DN A7 DN1
A8 SBU1
VBUS A4 B9 VBUS
VBUS B4 A9 VBUS
CC2 B5 USB_CC2
DP2 B6 USB_DP
DN2 B7 USB_DN
SBU2 B8
GND A1 B12
GND B1 A12
SHELL_GND SH1
SHELL_GND SH2
SHELL_GND SH3
SHELL_GND SH4
USB C 16P
USB_DN 1 CR13 2
USB_DP 1 CR14 2
GND
STM32H7 Board[1A] USB2_TX PB10
STM32H7 Board[1A] USB2_RX PB11
J22
USB2_CC1 A5 CC1
USB2_DP A6 DP1
USB2_DN A7 DN1
A8 SBU1
VBUS A4 B9 VBUS2
VBUS B4 A9 VBUS2
CC2 B5 USB2_CC2
DP2 B6 USB2_DP
DN2 B7 USB2_DN
SBU2 B8
GND A1 B12
GND B1 A12
SHELL_GND SH1
SHELL_GND SH2
SHELL_GND SH3
SHELL_GND SH4
USB C 16P
VBUS2 R50 22.1k
R51 47.5k
C39 1uF 0603
U11
VDD 6
VIO 5 3V3
7 VREGIN
RST 9 R49 1k 3V3
8 VBUS
USB2_DP 3 D+
USB2_DN 4 D-
RI/CLK 1
SUSPEND 17
SUSPENDB 15
18 CTS
19 RTS
22 DSR
24 DCD
23 DTR
GPIO.0/TXT 14
GPIO.1/RXT 13
GPIO.2/RS485 12
GPIO.3/WAKEUP 11
PB10 20 RXD
PB11 21 TXD
GND 2
GND 25
CP2102N-A02-GQFN24R
3V3
C37 100 nF 0402
C38 4.7uF X7R
USB2_CC1 R52 5.1k
USB2_CC2 R53 5.1k
5V D13 1N5819HW-7-F VBUS2
SB2
USB2_DN 1 CR15 2
USB2_DP 1 CR16 2
Title PI_CON
Size B
Number 04
Revision 01
Date: 6/19/2026
File: C:\Work\Altium\EDR\PI_CON.SchDoc
Sheet of Neural Nexus
Drawn By: Dulana Pitiwaduge